Real-time Fluorescence PCR Instrument

User Manual

Product information

Product: Real-time Fluorescence PCR Instrument

Specification: 8 Wells Portable Type

Manufacturer

Version information

This manual may be updated without further notice.

Version number: 1.2

Release date: 2023.12

International Symbols Description

Symbol	Description	Symbol	Description
	Keep away from open flame		**Keep dry**
	This side up		**Handle with care**
	Fragile		**Keep away from sunlight**

Table 1. Symbol Description

Technical Support and Warranty

Maintenance

This product is guaranteed for one year. Please keep the packing box for transportation, storage, or for use when product needs to be shipped for maintenance.

We are responsible for the safety, reliability and performance of this product under the following conditions:

- The instrument is assembled, upgraded, or maintained by an authorized party.
- Relevant electrical equipment complies with national standard.
- This product is used in accordance with this instruction manual.

If the product is in the following circumstances, which is not covered by the warranty, the consumer may need to pay the full amount for maintenance costs.

- The product is damaged due to the following reasons:
  - ◆ Improper operation
  - ◆ Improper connection with other devices
  - ◆ Accidents
- Modification of products without authorization
- Unrecognized product serial number

Technical Support

If technical support is needed, please contact your local distributor.

Please identify the following information:

- Type of malfunction or damage
- Product model and serial number

Safety Notice

Warning messages

The terms “Danger”, “Warning”, “Caution”, and “Attention” used in the manual indicate hazards and remind users to pay attention to the process of operation according to the degree of hazards. A hazard is a source of potential harm or adverse effects to the environment or user.

	Attention	Provide users with useful information and tips to ensure maximum performance of this product.
	Caution	Refers to a potentially hazardous situation which, if not avoided, may result in minor or moderate injury.
	Warning	Refers to a hazardous situation which, if not avoided, could result in death or serious injury.
	Danger	Refers to an immediate hazard which, if not avoided, will result in death or serious injury.

Safety Information

	Warning	1. Do not let the lens or image processing unit get wet and avoid the splashing reagents onto the device. 2. Make sure that the ambient temperature during operation is kept at 10- 35℃. 3. Please treat the amplification products of this instrument as biohazards and dispose of it properly. 4. To safely use this product, the hospital must provide a 110V/220V, 50/60Hz mains power output socket with standard protective grounding. 5. Do not plug or unplug the cable until the entire system is fully switched off.
	Caution	1. Without the permission or written authorization of the manufacturer, the user shall not disassemble, modify or use various accessories that do not meet the manufacturer's regulations. 2. Please switch off and unplug the device before cleaning it. 3. Do not disassemble the instrument. Doing so may cause an electric shock. 4. Please follow the instruction for cleaning the instrument. 5. To prevent damaging the instrument, please carefully follow the instructions described in this manual. 6. This instrument should be regularly calibrated and maintained by qualified technical personnel.

Table of contents

Chapter 1 Introduction ..... - 6 -

1.1 Intended use ..... - 6 -

1.2 Product features ..... - 6 -

1.3 Introduction of Real-time Fluorescence PCR Instrument ..... - 6 -

1.3.1 Components ..... - 6 -

1.3.2 Instrument parameters and characteristics ..... - 7 -

1.4 Unpacking and shipment ..... - 8 -

Chapter 2 Instrument Installation ..... - 9 -

2.1 Unpacking ..... - 9 -

2.1.1 Packing contents ..... - 9 -

2.1.2 Packaging protection ..... - 9 -

2.1.3 Unpacking guidelines ..... - 10 -

2.2 External connection ..... - 10 -

2.3 Installation ..... - 10 -

Chapter 3 Getting started ..... - 11 -

3.1 Inspection before starting up ..... - 11 -

3.2 Starting the instrument ..... - 11 -

Chapter 4 Software of the PCR instrument ..... - 12 -

4.1 Introduction ..... - 12 -

4.2 Function overview ..... - 12 -

4.2.1 Main interface ..... - 12 -

4.2.2 Previous data review ..... - 12 -

4.2.3 System Management ..... - 13 -

4.3 Instrument setting ..... - 13 -

4.3.1 WiFi connection setting ..... - 13 -

4.3.2 Wired connection setting ..... - 14 -

4.3.3 Date and time settings ..... - 15 -

4.3.4 Instrument maintenance ..... - 15 -

4.4 Testing ..... - 16 -

4.4.1 Setting the sample name and test items ..... - 16 -

4.4.2 Test running ..... - 18 -

4.4.3 Checking results ..... - 19 -

4.5 Previous data ..... - 21 -

4.5.1 Previous data review ..... - 21 -
4.5.2 Data export ..... - 21 -
4.5.3 Checking amplification curves ..... - 22 -
4.5.4 Query ..... - 22 -
Chapter 5 PC Software ..... - 24 -
5.1 Software installation ..... - 24 -
5.2 Connect instrument ..... - 25 -
5.3 Previous data review and Data update ..... - 25 -
5.4 Test report ..... - 26 -
5.5 Checking amplification curves ..... - 28 -
Chapter 6 Cleaning and Maintenance ..... - 29 -
6.1 Cleaning ..... - 29 -
6.1.1 Instrument surface cleaning ..... - 29 -
6.1.2 Sample well cleaning ..... - 29 -
6.2 Maintenance ..... - 29 -
Chapter 7 Troubleshooting ..... - 30 -
7.1 Troubleshooting Guides ..... - 30 -
Chapter 8 Technical support ..... - 31 -

Chapter 1 Introduction

This manual provides general information and operating instructions of the real-time PCR instrument. This chapter provides an overview of hardware, software, and technical support information.

Summary of operating, applications and features:

The PCR analyzer is designed for performing real time polymerase chain reaction (RT-PCR). It supports single or multiplex RT-PCR detection of interested targets in the clinical specimen. A color touchscreen and intuitive software makes the analyzer user friendly. The closed reaction system of the analyzer can reduce contamination and increase productivity. The software can collect and analyze data in real time, enables users to monitor data and print testing reports.

1.1 Intended use

This instrument is intended to detect DNA/RNA of targeting pathogens using fluorescence-based reagents.

1.2 Product features

The real-time PCR instrument provides rapid and accurate detection of multiple genes. Testing report can be monitored and printed.

1.3 Introduction of Real-time Fluorescence PCR Instrument

1.3.1 Components

This product is composed of heating and cooling system, temperature monitoring system, fluorescence detection system, cooling system, and power cord, as shown in Figure 1.

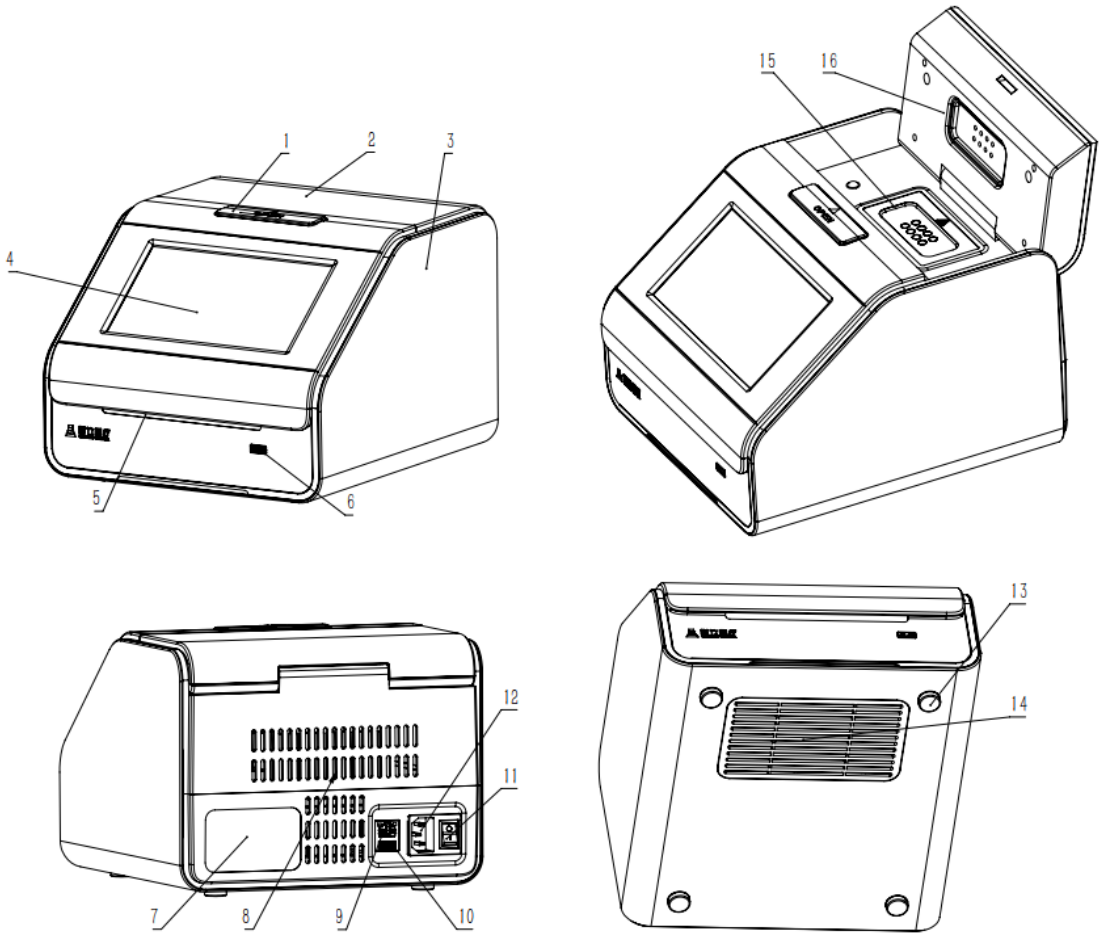

Fig.1 Components of the real-time fluorescence PCR instrument

1.Lid switch 2. Lid 3. Left panel 4. Screen 5. LED indicator 6. USB port 1
7. Label 8. Vent 9. Network port 10. USB port 2 11. Power switch 12. Power port
13. Case feet 14. Vent 15. Sample well 16. Heating lid

1.3.2 Instrument parameters and characteristics

1.3.2.1 General parameters of the instrument

►Instrument specification	
Dimensions:	268*236*188 mm
Weight:	4.5 kg
►Package specification	
Dimensions:	400*400*400 mm
Weight:	6.5 kg
►Power Supply	
Voltage:	AC 110V/220V
Frequency:	50-60Hz
►Connectivity	
Network connection:	TCP / IP, Ethernet

►**Operating Conditions**

Temperature:	10℃-35℃
Humidity:	10%-95%RH
Atmospheric pressure:	Standard atmospheric pressure
Altitude:	2000 meters
Storage:	Away from moisture
Noise:	50dB

1.3.2.2 Technical parameters of the instrument

Thermal parameters	
►Temperature accuracy	±0.25℃
►Temperature uniformity	±0.25℃
►Temperature control accuracy	±0.1℃
►Rate of Heating and Cooling	6℃/S
Optical parameters	
►FAM emission	520nm±10nm
►HEX emission	570nm±10nm
►FAM excitation	495nm±10nm
►HEX excitation	535nm±10nm
Interaction system parameters	
►Touch screen:	7-inch color touch screen
►USB port:	Connect USB flash drive and barcode scanner Prohibit：The USB port of this instrument cannot be used for charging electronic devices Note: Before connecting the USB flash drive to the instrument, please format the USB flash drive to FAT32, and confirm that there is enough space (1 GB recommended) in the USB.

1.4 Unpacking and shipment

Do not turn the instrument upside down. After taking out the instrument and accessories, keep the packaging materials for use in shipments for repairs. Check the accessories according to the packing list.

In addition, you should receive all the optional parts and accessories you ordered. If there is anything missing, please contact our local distributor. If damage occurs during transportation, keep the packing materials for inspection and contact your local distributor immediately. According to the packing list provided by the manufacturer, please carefully check if the instrument and accessories are correct, and if there is any damage. Please fill in the product warranty card carefully and send it Return to our company within one month of receiving the product.

Chapter 2 Instrument Installation

2.1 Unpacking

2.1.1 Packing contents

When you receive the real-time PCR instrument, please check that it includes the following items.

No.	Item	Specification	Quantity	Unit
1	Real-time Fluorescence PCR Instrument		1	Unit
2	Power cord		1	Piece
3	USB disk		1	Piece
4	Experiment operation guide		1	Sheet
5	User manual		1	Booklet
6	Pipette	100-1000μL	1	Unit
7	Pipette	2-20μL	1	Unit
8	Pipette tip	1mL	1	Box
9	Pipette tip	200μL	1	Box
10	Certificate		1	Sheet
11	Warranty card		1	Sheet
12	List of product contents		1	Sheet
13	QR code scanner		1	Unit

2.1.2 Packaging protection

This product is transported with well-designed carton packaging. The periphery and corners of the instrument are supported and fixed by the inner lining protective foam to prevent collision and vibration during transportation, as shown in Figure 2.

Fig2. Packing lining

2.1.3 Unpacking guidelines

Step 1: Use scissors or knife to open the packaging box.

Step 2: Remove the top layer of foam pad and take out the accessories.

Step 3: Take out the instrument and the power cord.

Step 4: Place the instrument in a suitable environment, plug in power cord and turn it on.

2.2 External connection

Power requirement: This instrument uses a power switch. The operating voltage is 110V/220V, and the frequency is 50/60HZ. Power line is a single phase 3-wire system and should be reliably grounded.

Warning	**Electricity must be grounded to avoid instrument from short circuits or causing an electric shock.**

2.3 Installation

1. The operating environment of the instrument should meet the environmental requirements of 2.2.1 Keep the instrument on a flat, stable, and dry surface.

2. Please use the power cord provided with the instrument. When connecting, set the power switch to the OFF position.

Warning	**If the power cord is broken, it must be replaced.** **Replacement must be replaced with a power cord of the same type and size.**

Chapter 3 Getting started

3.1 Inspection before starting up

Before switching on the power, the following contents should be confirmed first:

1. Whether the power supply conforms to the system requirements.

2. Confirm that the power plug is correctly and reliably inserted into the power socket.

3. Whether the surrounding working environment and the placement conditions of the instruments meet the requirements.

3.2 Starting the instrument

Connect the power cord to the instrument, plug it into a power outlet, and set the power switch to "-". The instrument will enter the booting interface and do self-testing. After booting up, the operating system interface appears, as shown in Figure 3.

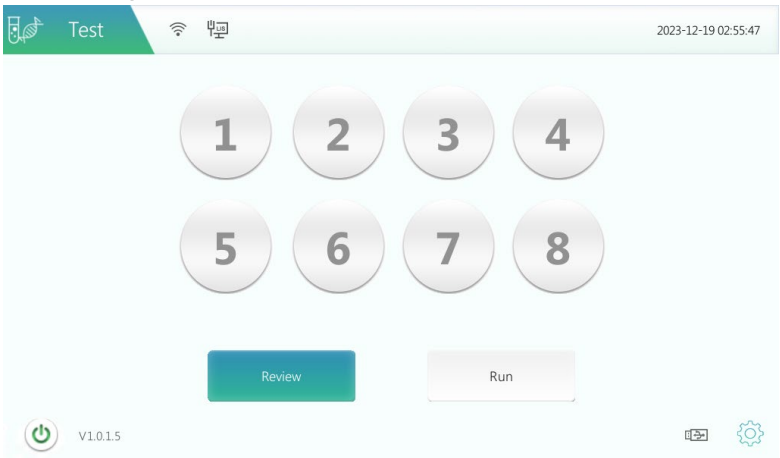

Fig3. Main interface

Chapter 4 Software of the PCR instrument

4.1 Introduction

The instrument software can not only help users complete the PCR detection, but also automatically complete the collection, analysis and processing of data. More functions include previous data review, data export, amplification curve checking and other system function settings, etc.

4.2 Function overview

4.2.1 Main interface

The main interface will display top bar, middle bar and bottom bar. The upper left corner of the top bar displays the name of the operation interface, and the upper right corner displays the time and date. The middle bar displays the specific operation functions; the bottom bar displays the power button and software version on the left and the Logo on the right, as shown in Figure 4.

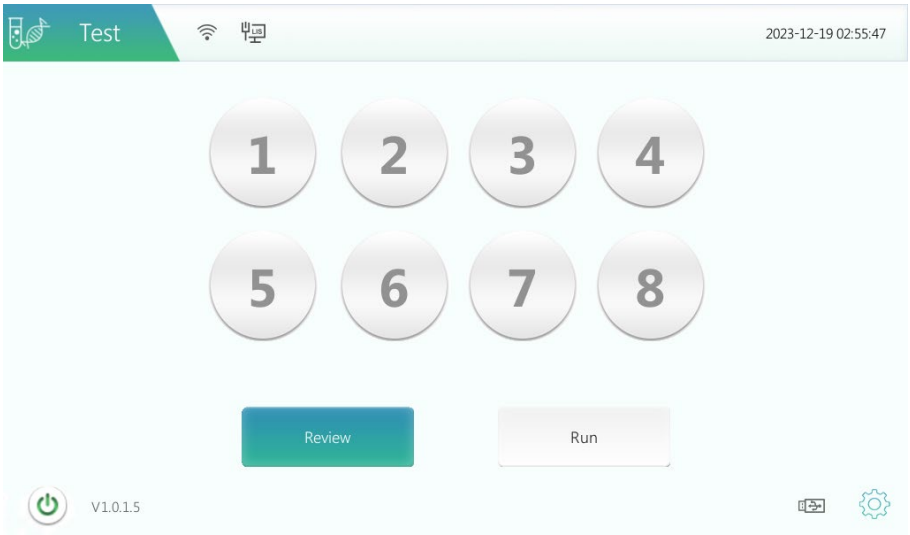

Fig 4. Main interface

4.2.2 Previous data review

Click the “Review” button, and enter the previous data interface, as shown in Figure 5. Click the top “Return” icon to return to the main interface.

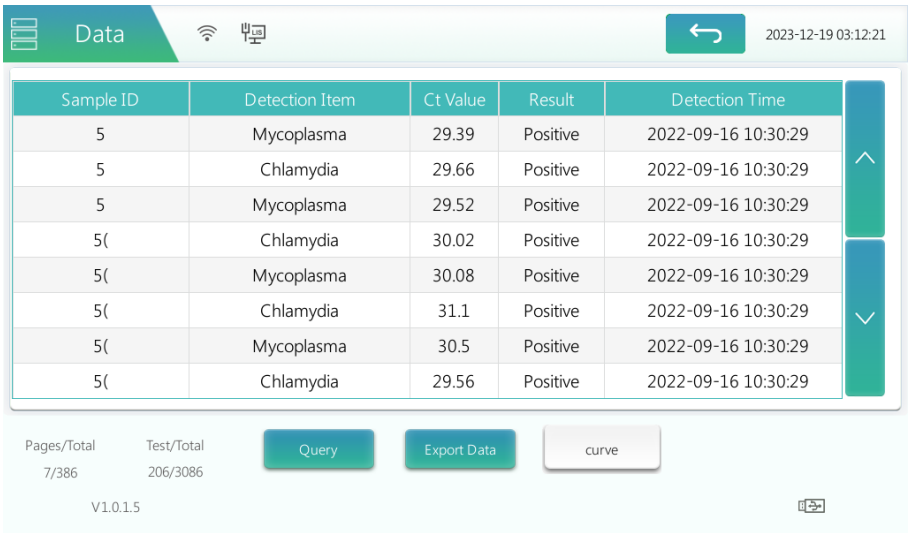

Fig5. Review interface

4.2.3 System Management

Long press the lower right corner Logo icon of the main interface for 2 seconds to enter the system setting interface, as shown in Figure 6. In the system setting interface, you can check the device information, adjust date and time, switch language, set touch sound, adjust screen brightness, import and export data, export log or import script, set the printer, connect network, enter the system maintenance interface or factory service interface etc. Click “Return” button to return to the main interface.

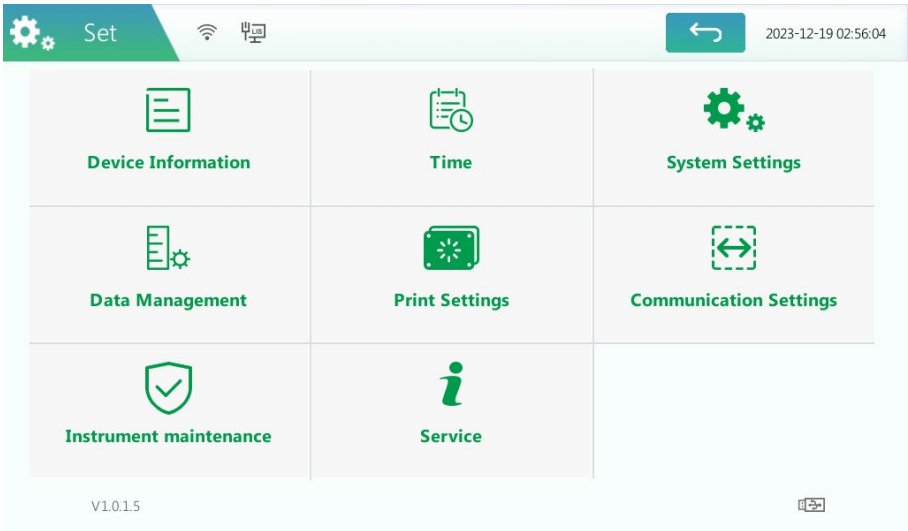

Fig6. Setting interface

4.3 Instrument setting

4.3.1 WiFi connection setting

Enter the system setting interface as shown in Figure 6, and click “Communication Settings” button, enter the interface shown in Figure 7. While using WiFi network, tick the right bottom circle first. Then click the “Scan WiFi” button, select the right WiFi name, enter the right WiFi password in the password box and then click the “Connect” button to connect to the WiFi. When connected, the WiFi icon will appear

in the top right corner. If don't, please check the password, or the WiFi signal is strong enough.

After connecting to WiFi, you can click "WiFi Details" to get the WiFi IP address. Please check Chapter 5 for more detailed information regarding the software.

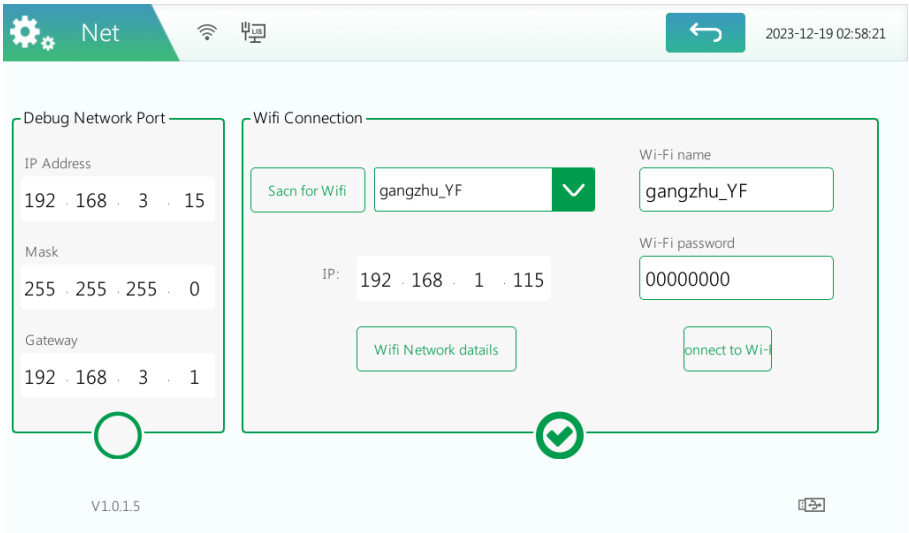

Fig7. Network connecting interface

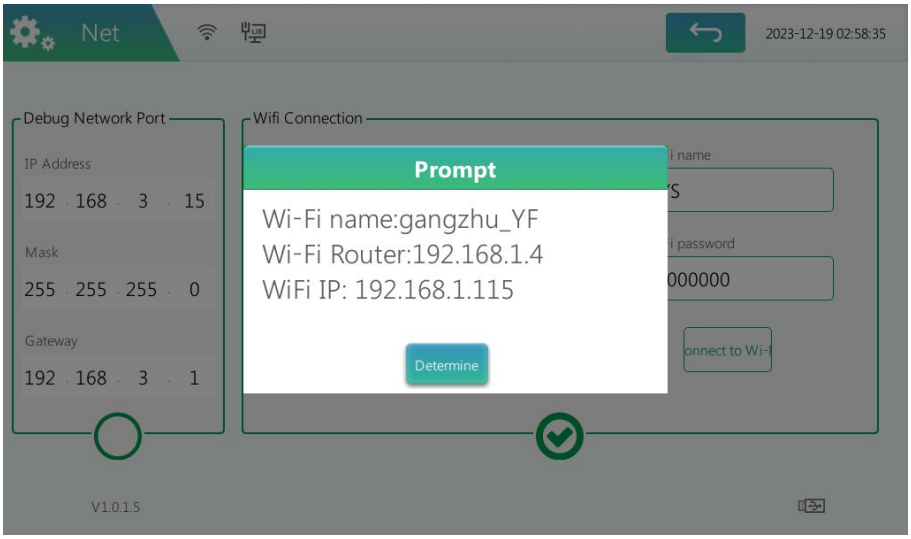

Fig. 8 WiFi details

4.3.2 Wired connection setting

Enter the system setting interface as shown in Figure 6, and click "Communication Settings" button, enter the interface shown in Figure 7. While using wired network, tick the left bottom circle first. Then set the right router address in the left information box, as shown in Figure 9.

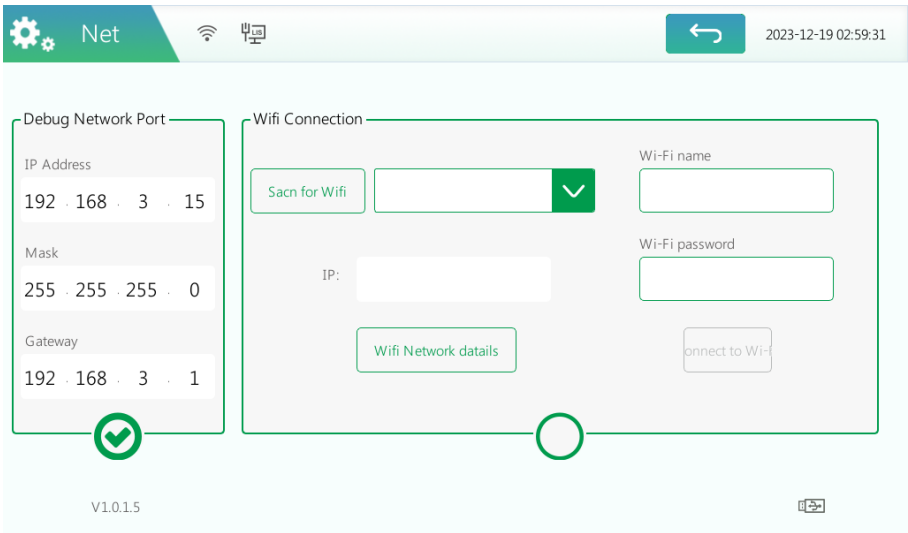

Fig. 9 Wired network setting

4.3.3 Date and time settings

Enter the system setting interface as shown in Figure 6, and click “Time” button, enter the interface shown in Figure 10. In the date & time setting box, you can use “Select” button to set the right date and time. Also, you set the display date format which you like in the date format setting box.

When the settings are completed, click “Return” button to exit. If the date & time information has been changed, it will pop up a message box to let you confirm to change or not. Choose “Yes”, it will save the change, otherwise, the change will not be saved.

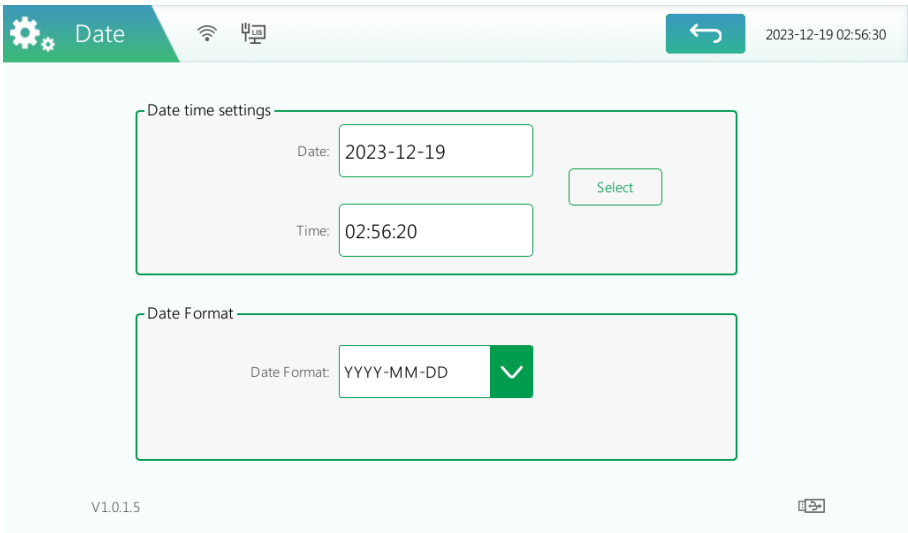

Fig.10 Date and time interface

4.3.4 Instrument maintenance

Enter the system setting interface as shown in Figure 6, and click “Instrument maintenance” button, enter the password input interface shown in Figure 11. Input the defined password “12345” to get in the instrument maintenance interface as shown in Figure 12. You can clear the instrument database or choose the test items for showing here.

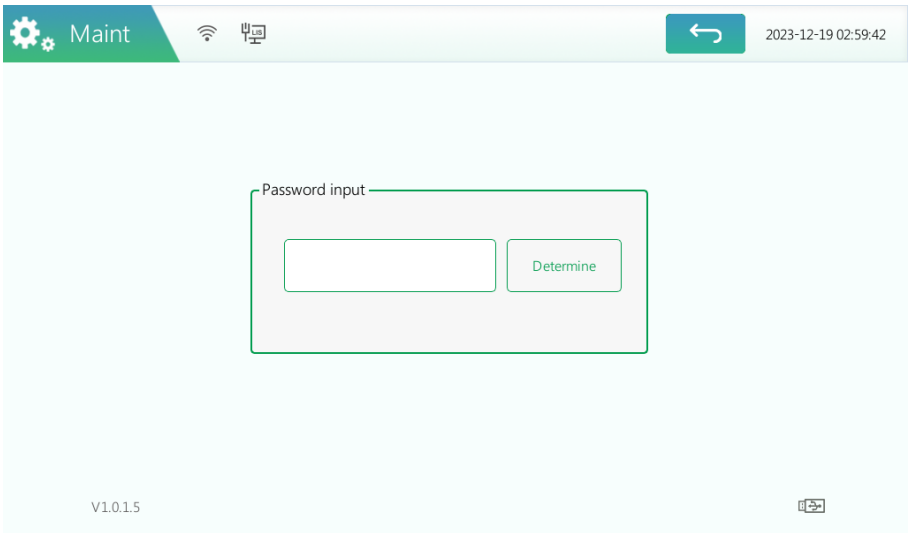

Fig.11 Password input interface

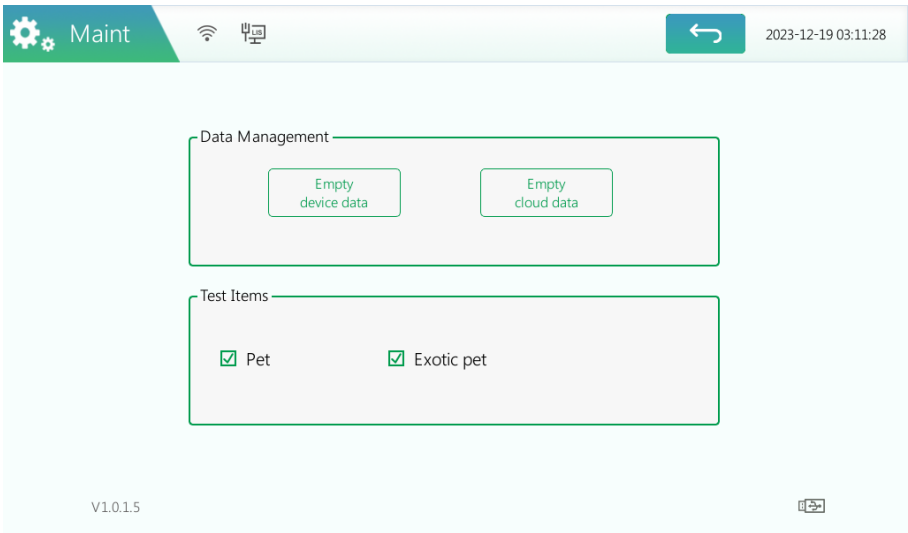

Fig.12 Instrument maintenance interface

4.4 Testing

This section introduces the real-time PCR software and how to complete a PCR test, including sample information settings, data collection, and data management.

4.4.1 Setting the sample name and test items

Turn on the machine and enter the main interface as shown in Figure 13. Press the “OPEN” button of the instrument to open the lid and put the test reagent tubes into the sample wells. Then select the corresponding sample wells in the main interface of the instrument (For example, if the test reagent is placed in the No.1 sample well of the instrument, click the No.1 position button on the main interface.) to get in the sample setting interface as shown in Figure 14. Here you can set the test sample name in the “Sample ID” box and test items in the “Reagent ID” box. You can use barcode scanner or keyboard to input the reagent ID. When the settings is completed, click “Save” button to save it and return to the main interface as shown in Figure 15.

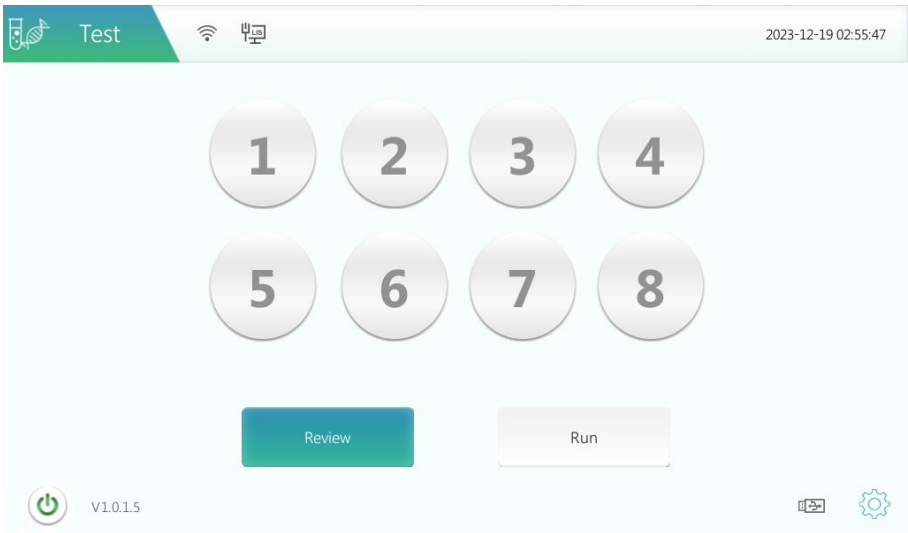

Fig.13 Main interface

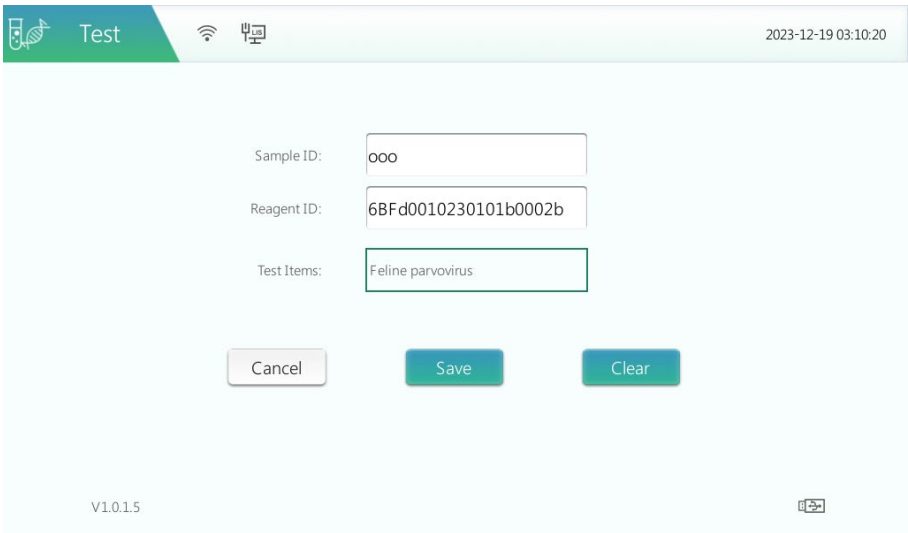

Fig.14 Sample setting interface

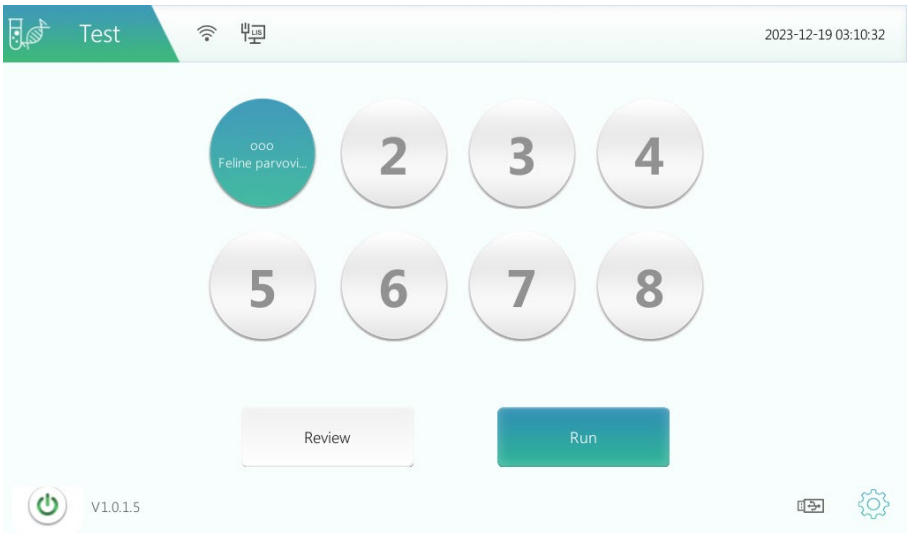

Fig.15 Main interface after setting sample information

Repeat the above steps, and you can set other sample information. Up to 8 sample information (All of 8 sample wells) can be set at the same time as shown in Figure 16.

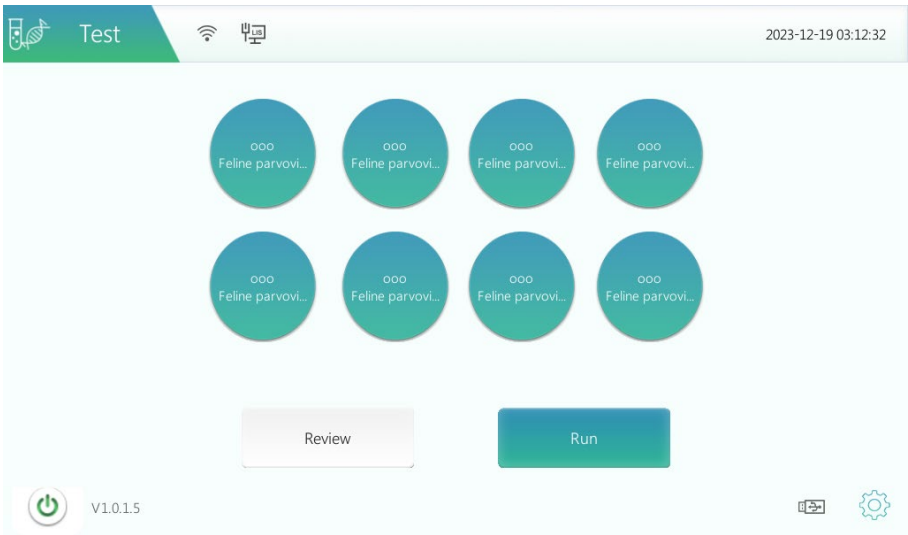

Fig.16 Main interface after 8 sample information being set

4.4.2 Test running

After setting the sample information, click “Run” button to start the test, as shown in Figure17.

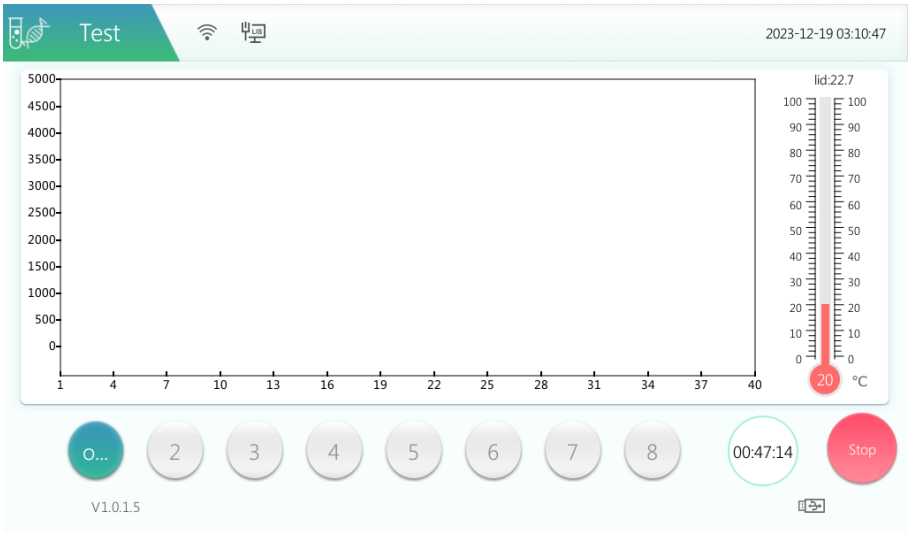

Fig.17 Running test interface

The middle part of the functional area displays the amplification curve of real-time PCR, the position of the sample well, and the operating temperature of the instrument respectively. The lower part of the functional area displays the sample well position button, the countdown timer, and the stop button.

Click the thermometer area switching to display the current testing information, as shown in Figure 18. this area displays the amplification curve with indicated color, testing items, and sample well location information. Click the sample well button in the lower part of the selection function area to display the corresponding real time information of the selected sample well.

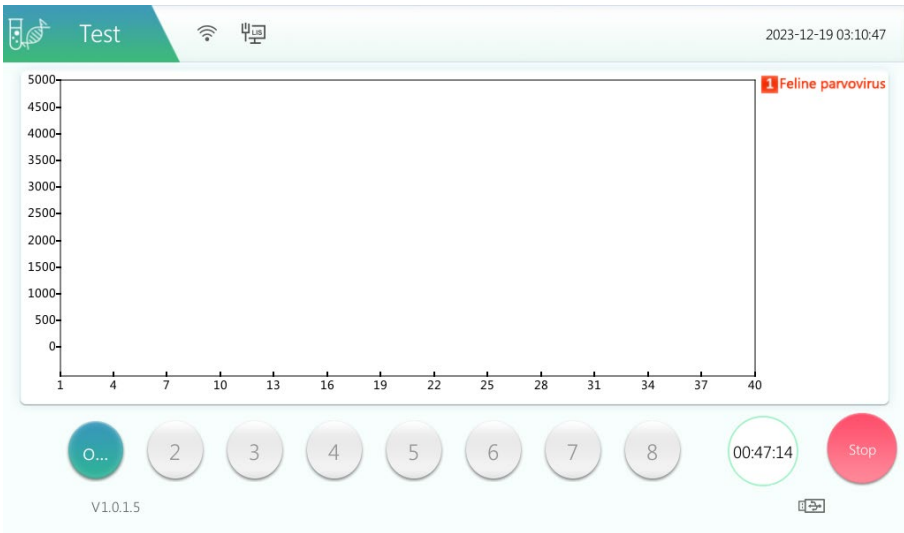

Fig.18 Running sample test information display

If you want to terminate the testing before it is completed, please click “Stop”. Then, it will pop up a notice box as shown in Fig 19 on the interface to let you choose whether stop the test or not. You can choose "Yes" to stop the test, or "No" to continue the test. If you stop the test before it is completed, the experiment data will not be saved and most of the time the reagent will be invalid. However, if it has just begun within 1 minute, the reagent will remain valid.

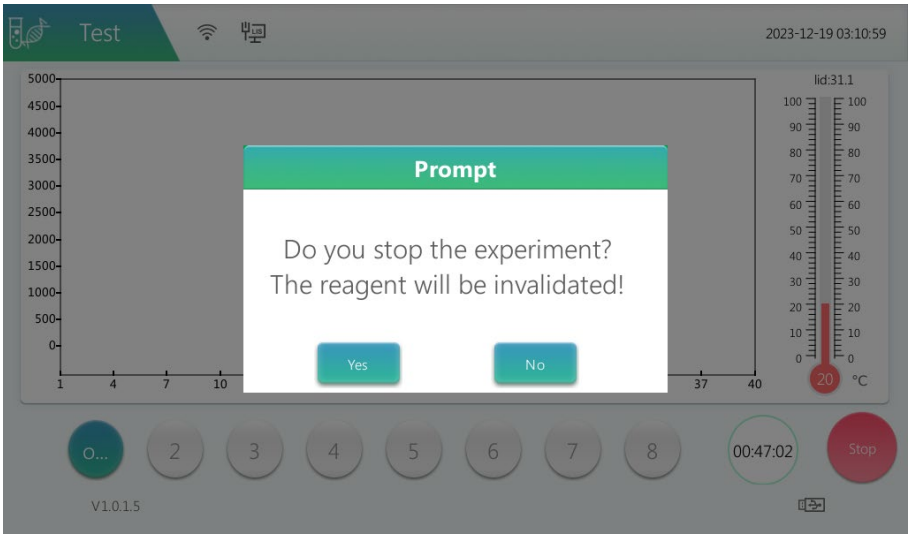

Fig.19 Stop testing

4.4.3 Checking results

When the test is completed, it will pop up a finished notice box on the interface (Fig.20); click the finished button to get in checking results status (Fig.21). The data were processed by the software. Click “View Results” to check the results of the sample well, as shown in Figure 22. You can click “Return” button to return to the main interface for the next experiment.

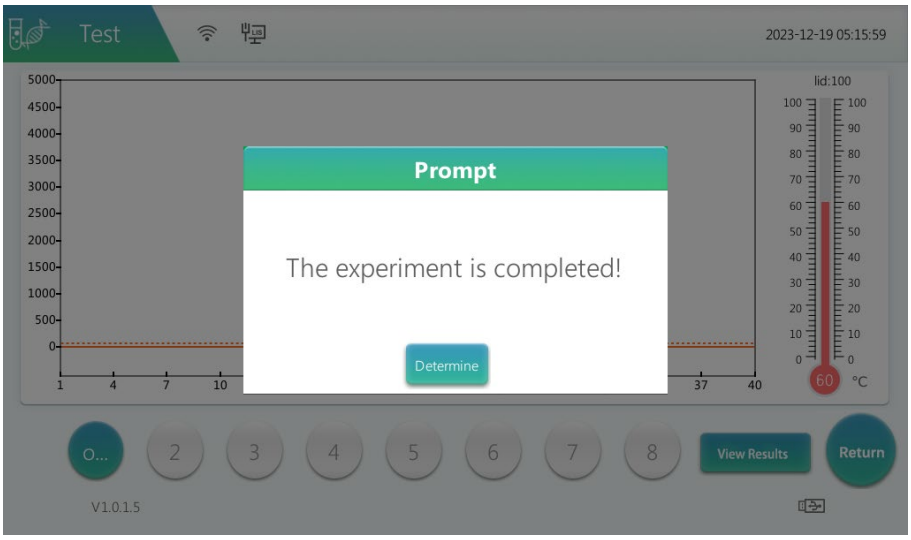

Fig.20 Completed testing interface

Fig.21 Completed testing interface

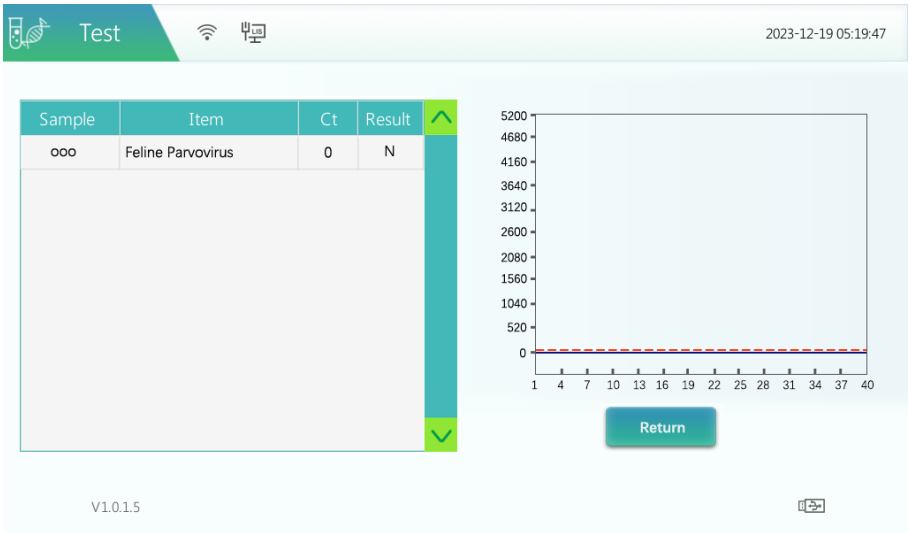

Fig.22 View results interface

4.5 Previous data

4.5.1 Previous data review

In the main interface, while the “Review” button is colored (by all the sample wells with no setting information), it can be selected to get into the previous data interface, as shown in Figure 23-24.

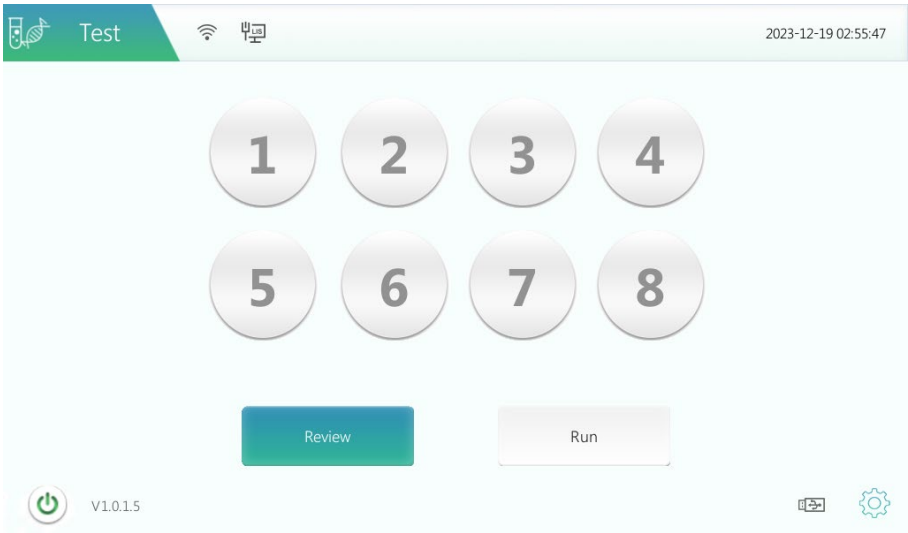

Fig.23 Main interface

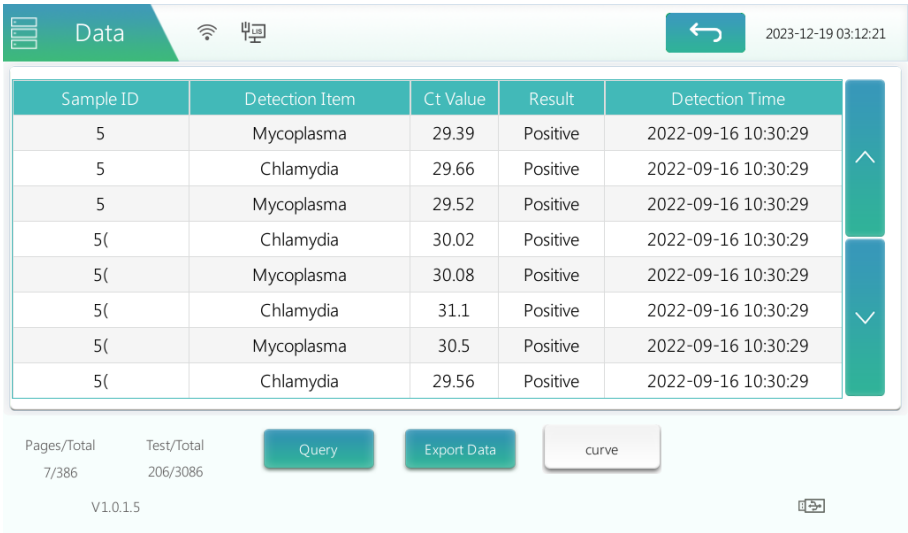

Fig.24 Previous data interface

Here you can Export Data by using an FAT32 format flash drive to export the data file (Appdatabase.db). Use the “Query” function to search previous testing data. Also check the selected test amplification curves by click the “Curve” button.

4.5.2 Data export

Prepare an FAT32 format flash drive, then plug it in the PCR instrument. After the USB icon appears, click “Export Data” to export the data file (Appdatabase.db) to the flash drive. When it is completed, the complete notice box will pop up, as shown in Figure 25.

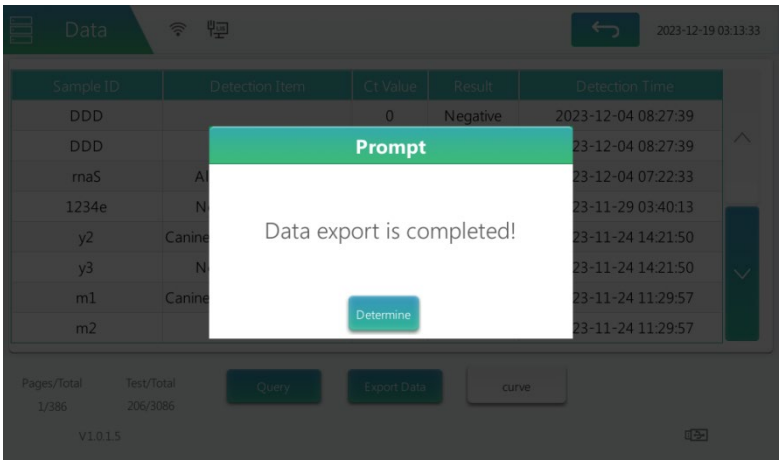

Fig.25 Export data completed

4.5.3 Checking amplification curves

Select one of test items which you want to check the amplification curve, click the "Curve" button. The interface will show all the test items' curves and results in the test selected at the same time (Fig.26). You can check each test item's amplification curve by selectively choosing the sample well and "FAM/VIC" icon.

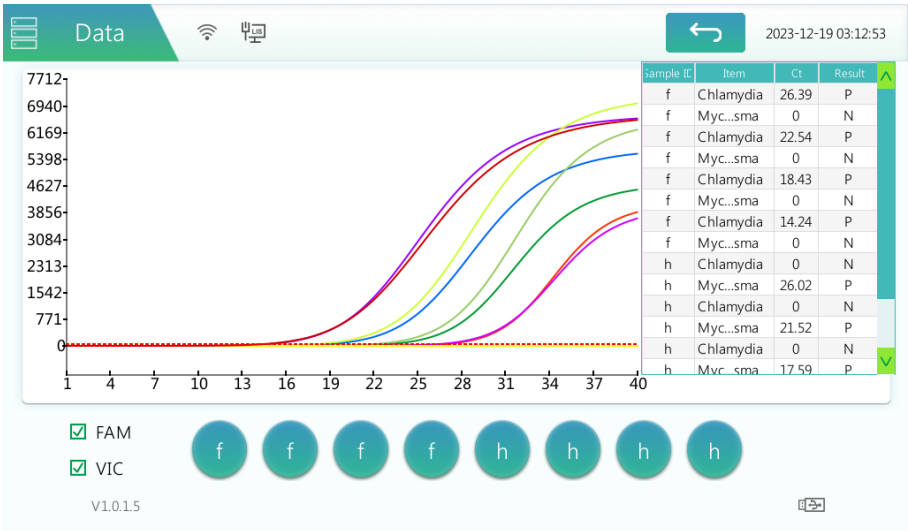

Fig.26 Review test amplification curves

4.5.4 Query

In the previous data interface, click the "Query" button to get in the history searching interface, as shown in Figure 27. Using the right searching information, you can get the results which you want to check more easily and quickly.

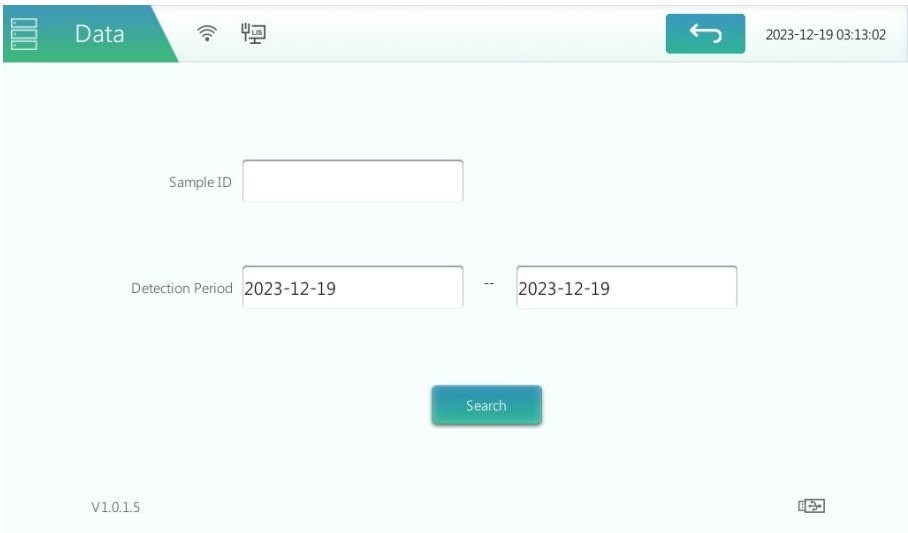

Fig.27 History searching interface

Chapter 5 PC Software

5.1 Software installation

Find the USB flash drive in the accessory of the instrument. Unzip the compressed install package from the flash drive to install the PCR reporting system software in the computer with Windows system (Win 7 above). The reporting system software installation folder contents are shown as Figure 28. Create a shortcut of the application to the desktop can make it use more easily.

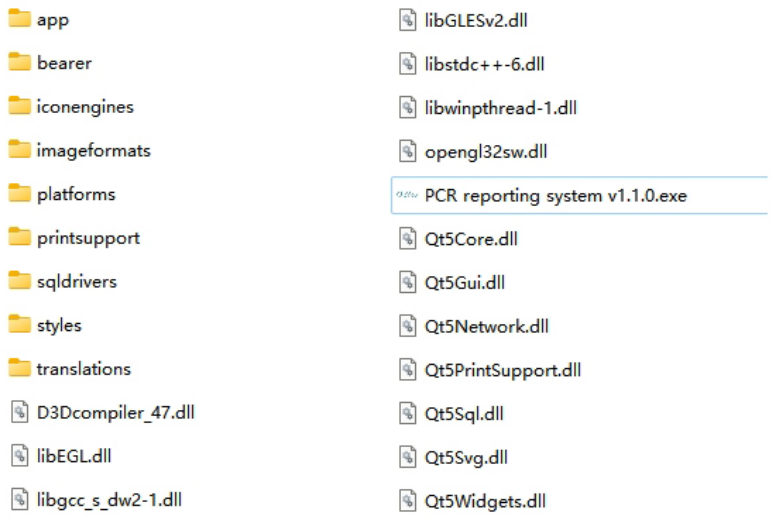

Fig.28 PC software folder

Double click on the application icon to open the reporting system software, as shown in Figure 29. The software navigation column lists on the left. Previous settings maybe set in the System settings and Template edit before using the reporting system.

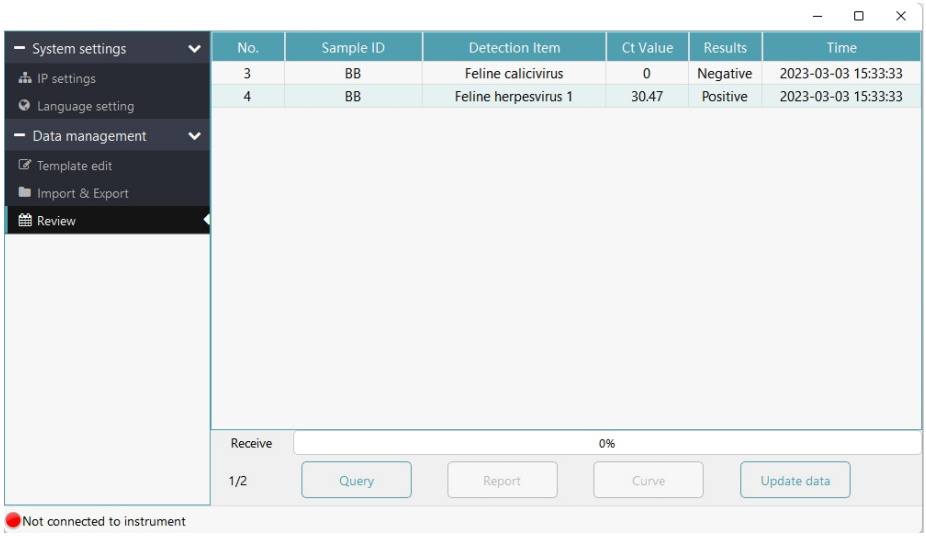

Fig.29 PC software interface

5.2 Connect instrument

While use network to transfer data from the PCR instrument, make sure the network used by the computer and the PCR instrument should be under the same network router. First, turn on the PCR instrument and obtain the IP address (ether use WiFi connection or wired connection, refer to 4.3.1 and 4.3.2) of the PCR instrument. Then input the IP address in the IP setting box and click “Set & Connect” button to connect the reporting system software with the PCR instrument, as shown in Figure 30. If the connection is abnormal, please check the network environment or computer system network settings.

Fig.30 PC software connection

5.3 Previous data review and Data update

Click "Data management -- Review" to enter previous data interface, as shown in Figure 31.

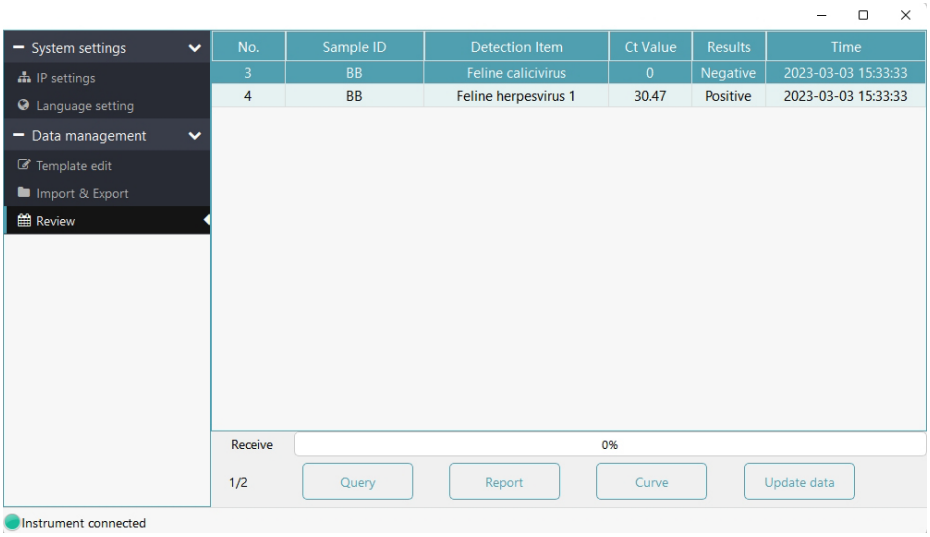

Fig.31 Previous data interface

After connected with the PCR instrument, you can update the data from the PCR instrument by click the "Update Database" button. When the process bar displays 100%, the data synchronization completed (Fig.32). If the data transfer fails, please check the computer firewall and network settings or WiFi signal.

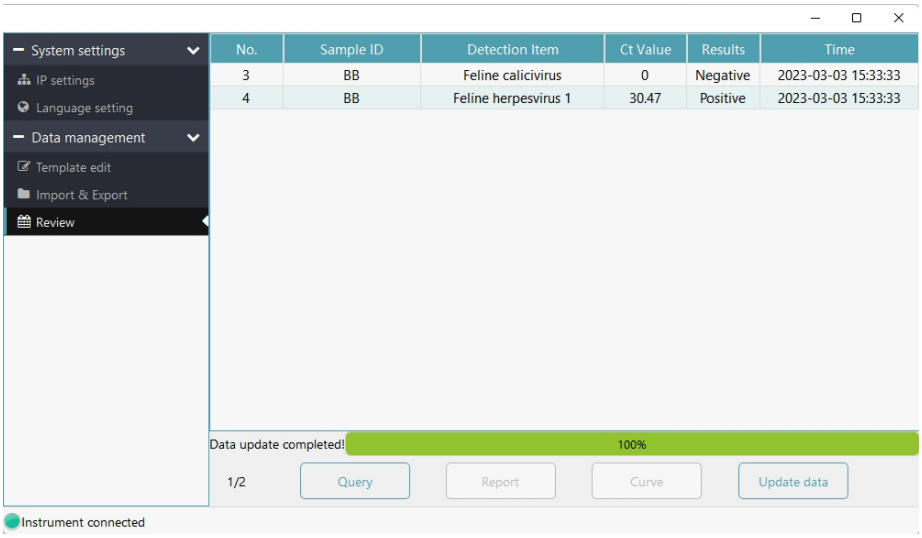

Fig.32 Data update

Note: While using WiFi network, please keep the PCR instrument and computer in a stable environment with strong WiFi network signal.

You can also use the flash drive to import data to update from the PCR instrument through the “Import & Export” function, as shown in Figure 33. The import & export data file name is “Appdatabase.db”.

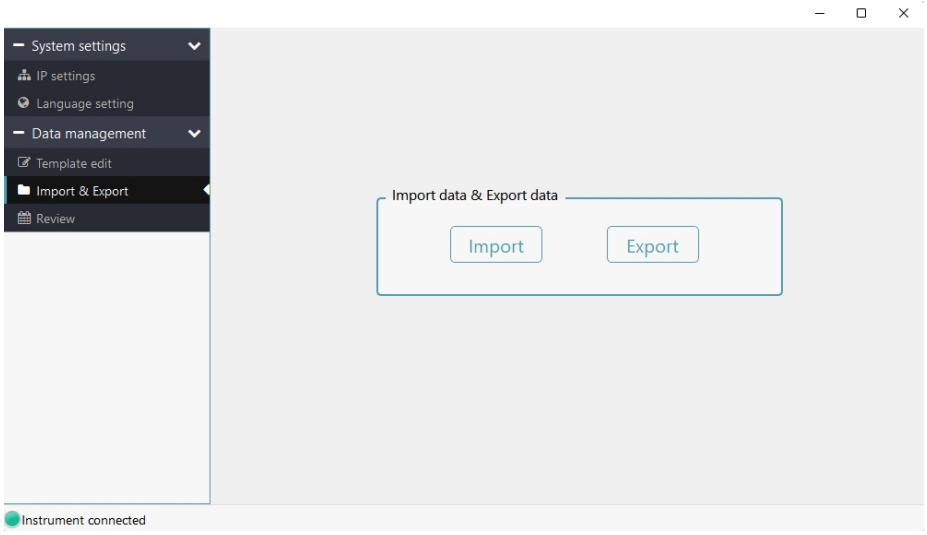

Fig.33 Data import and export

5.4 Test report

First time for using the reporting system, you should click “Template edit” to set the report template information as shown in Figure 34.

In the previous data interface, select the test item and click "Report" to enter the report editing interface (Fig.35). The information editing section is on the left, and the report preview section is on the right. After editing the information, click “Print” to enter the preview interface as shown in Figure 36. Then, click the top printer icon to complete the printing.

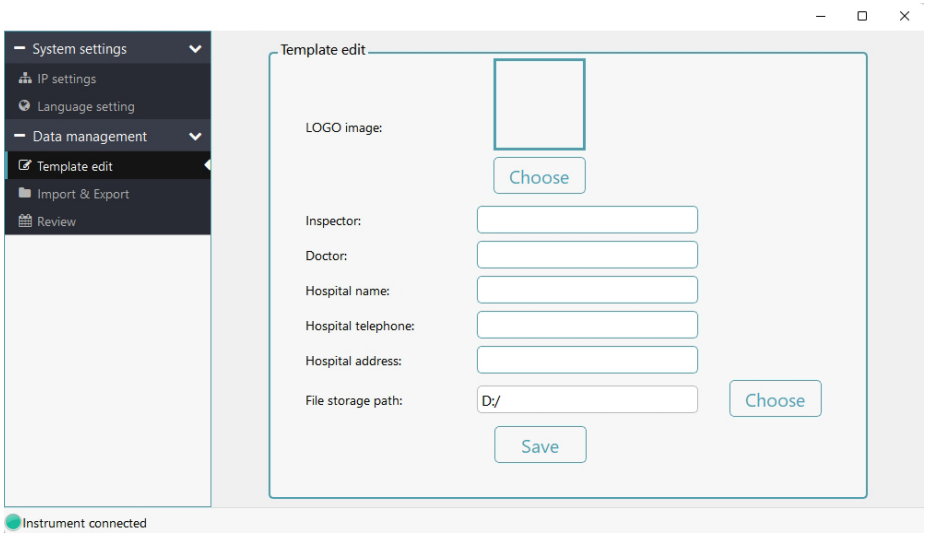

Fig.34 Report template setting

Fig.35 Report interface

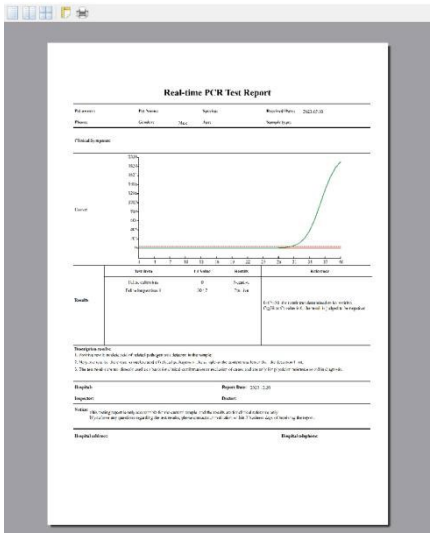

Fig.36 Printing preview

You can click "Export Report" button on the report editing interface to save the PDF reporting file to the path being set in “Template edit” (Fig.37).

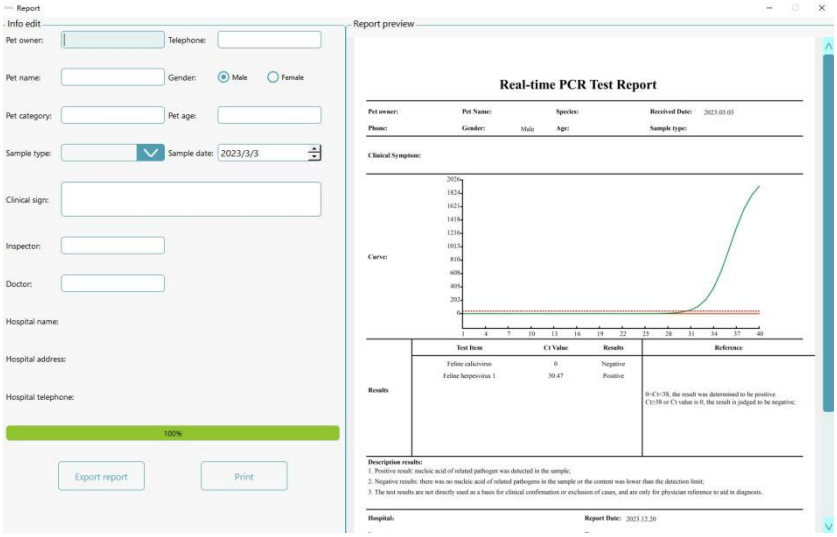

Fig.37 Save PDF file of test report

5.5 Checking amplification curves

In the previous data interface, select one of the test items, then click “Curve” to enter the amplification curve checking interface, as shown in Figure 38. The interface will show all the test items’ curves and results in the test selected at the same time. You can check each test item’s amplification curve by selectively choosing the sample well and “FAM/VIC” icon.

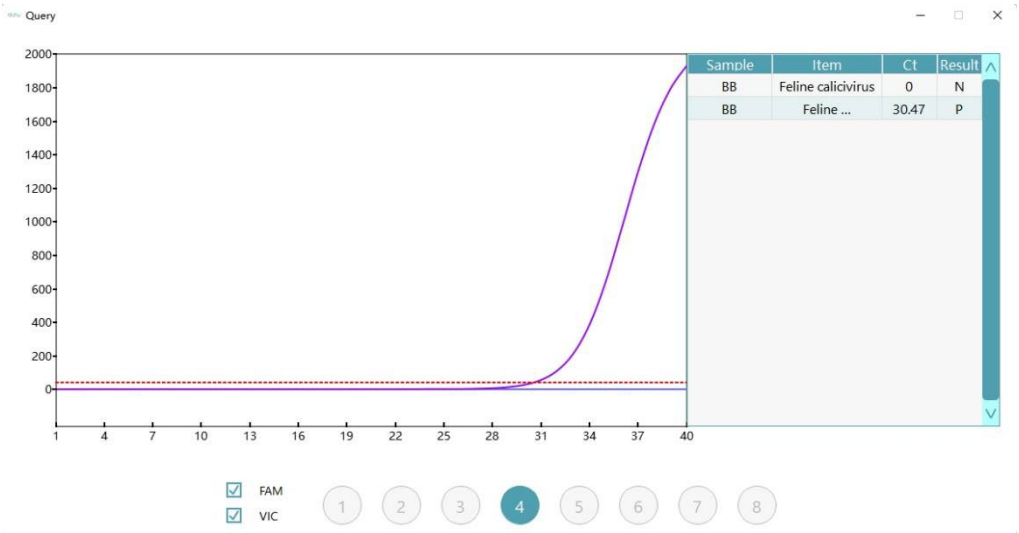

Fig.38 Amplification curve interface

Chapter 6 Cleaning and Maintenance

6.1 Cleaning

6.1.1 Instrument surface cleaning

The surface of the instrument should be cleaned regularly using a slightly moist soft cloth with water, and the instrument should be wiped dry after cleaning.

6.1.2 Sample well cleaning

1. If the sample well is contaminated with dust or impurities, it will affect PCR amplification and fluorescence detection. Therefore, it should be cleaned regularly, usually once every three months, and can be gently blown with a blower.

2. In order to prevent dust from entering the sample well, the lid should be closed when the instrument is not in use.

3. If any reagent enters the sample well, wipe it clean using a dust-free soft cloth with 70% alcohol.

!	Notice	Switch off and unplug the instrument before cleaning. Do not pour liquids into the reaction module or inside the instrument. Do not scrub the instrument with strong corrosive or organic solvents.

6.2 Maintenance

1. Do not switch the instrument on and off frequently.

2. Do not turn off the power immediately after testing, keep it in standby mode for 10 minutes (the internal fan of the instrument is still working continuously during this period). Turn off the power after the module temperature drops to room temperature.

3. Please use the power cord provided by the manufacturer.

4. Do not use hot water bath or low temperature insulation (such as 4° C) on the instrument.

5. Do not disassemble the instrument. If any problem occurs, please contact your local distributor.

Chapter 7 Troubleshooting

7.1 Troubleshooting Guides

No.	Problem	Possible causes	Solution
1	After booting, the status LED is normal, but the screen is black	Abnormal connection	Check if the circuit is stable, if the voltage is normal, and if the power interface is tightly plugged in
2	Touch screen is unresponsive	Faulty screen or poor cable contact	Contact maintenance personnel
3	Fan does not work	Internal cable problem or fan failure	Shut down the instrument, and contact maintenance personnel
4	There is no response when the USB flash drive is inserted into the USB port	The format of USB flash drive is not compatible	Check if the format of USB flash drive is FAT32
5	Wi-Fi network is not detected	The wireless frequency of the WiFi network does not match	Check if the network is 2.4G WiFi network
6	Network error, please check the network	The device is not connected to the Internet, or the connected network signal is unstable, or the router is faulty	Check the network environment
7	When uploading data, it prompts "unknown error"	The device is not connected to the Internet, or the connected network signal is unstable, or the router is faulty	Check the network environment
8	Heating abnormal fault prompt	Abnormal heating	Contact maintenance personnel
9	Fault prompt for abnormal temperature drop of the instrument	Abnormal cooling of the instrument	Contact maintenance personnel
10	Heating lid sensor abnormal fault prompt	Abnormal heating lid sensor	Contact maintenance personnel
11	Abnormal fault prompt of the heat sink sensor	Abnormal heat sink sensor	Contact maintenance personnel
12	Heat sink over temperature fault prompt	Abnormal heat sink	Contact maintenance personnel

Chapter 8 Technical support

We guarantee that the real-time fluorescence PCR instrument has passed a comprehensive test and meets the requirements in the instruction manual.

The use of this instrument must comply with the instructions and safety warnings given in this user manual, otherwise it will not be covered by the guarantee.

Software: Any software related to this product is provided to customers free of charge as a service, and the software is a necessary tool to run the instrument.

Responsibility: We are not responsible for direct, indirect, or incidental damage caused by failure to comply with the instructions given in the user manual or incorrect use of the real-time PCR instrument. Only our technicians or their agencies can inspect or provide any parts of the instrument. We are not responsible for any direct, indirect or incidental damage caused by the user's random disassembly and replacement of parts.

The testing results are for in vitro veterinary diagnosis only and cannot be used as the full or direct basis for decision-making on clinical medication, surgery, and prognosis. The veterinarian should combine the judgment of the clinical symptoms of the pet with other diagnostic results.

Factors may affect the results during PCR testing include:

Environmental factors such as aerosol pollution of the previous testing items, unstable voltage, etc. Human operation factors such as extraction errors, failure to strictly follow the requirements of the instruction manual, etc.

Equipment factors such as vibration, improper use, etc.

Kit storage factors such as high temperature, high humidity, strong direct light, unqualified reagent quality, etc.

Test results may appear false negative, or false positive, or inconsistent with clinical symptoms due to the factors mentioned above. The test results are for in vitro veterinary diagnosis only. Please consult all other clinical symptoms and use other diagnostic methods, to ensure an accurate diagnosis.

Before using the real-time PCR instrument, please read this manual carefully to ensure that you can get the best testing results.

If you have any questions regarding the testing process, please contact our technical support.